

## Introduction for students: why Chapter 5 matters

Chapter 5 introduces the first active element of the course: the operational amplifier, or op amp. With two rules and the node equations of Chapter 4 you can design circuits that amplify, add, subtract and buffer signals. Chapter 5 runs from week 9, session 2, to the end of week 10, and its quiz is at the start of week 11.

### What you will learn

- **The ideal op amp:** with negative feedback and in its linear region, no current enters either input ( $i_p = i_n = 0$ ) and the two input voltages are equal ( $v_p = v_n$ ).
- **Saturation:** the output can never go beyond the supply voltages. Outside that range the two rules no longer hold.
- **The inverting amplifier:**  $v_o = -(R_f/R_s)v_s$ .
- **The summing amplifier:** several inputs, each scaled by  $R_f$  over its own input resistor, then added and inverted.
- **The noninverting amplifier:**  $v_o = (1 + R_f/R_s)v_g$ , and with  $R_f = 0$  the voltage follower, which copies a voltage without loading it.
- **The difference amplifier:**  $v_o = (R_b/R_a)(v_b - v_a)$  when the resistor ratios on the two inputs match.
- **Cascades:** the output of one stage drives the next, and each stage must stay inside its supplies.

### Later in this course

| Topic                               | Where Chapter 5 comes back                                                                                            |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Chapter 6: Inductors and capacitors | An op amp with a capacitor in the feedback path integrates its input; the same node equation with a capacitor current |
| Chapter 7: First-order circuits     | Op-amp circuits with a capacitor charge and discharge with a time constant, found with the methods of Chapter 7       |

### In your program

- **Electrical and Electronics:** amplifiers, filters, voltage regulators and the control loops of motor drives are built around op amps.
- **Computer, Communication and Telecom:** every analog signal that reaches a processor, from a microphone, an antenna or a sensor, passes through op-amp stages that scale and shift it for the analog-to-digital converter.
- **Biomedical:** an ECG amplifier is a difference amplifier: it amplifies the tiny difference between two electrodes and rejects what both pick up.
- **Mechanical and Industrial:** strain gauges, thermocouples and pressure sensors give millivolts; op amps turn them into volts a controller can read.
- **Surveying:** the sensors inside levels, total stations and GNSS receivers feed op-amp stages before their signals are digitised.

## As an engineer

- **Two rules, then node equations.** Check for negative feedback, set  $v_n = v_p$  and  $i_n = i_p = 0$ , and write KCL at the inverting input. Every circuit in this chapter yields to that recipe.
- **Always check the supplies.** An answer outside the supply range is not an answer: the op amp saturates, and the output sits at a supply voltage.
- **The inputs draw no current; the output can.** The output current comes from the supplies, which is why an op amp can drive a load that would collapse a divider.

## In real life

- **Monitoring a solar battery bank.** A divider scales 60 V down to 5 V for a display, but the display's own resistance pulls the reading down. A voltage follower between them fixes it (S7).
- **From a sensor to a microcontroller.** A temperature sensor gives 10 mV per °C; an op amp scales and shifts that signal so that the microcontroller's 0–5 V input range is used in full (S15, and Nilsson's Practical Perspective for this chapter).

## How to study this chapter, and where AI fits

- AI tools are allowed on the practice sheet. Use them to check your work, not to replace it.
- In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused about a third of its wrong answers, and misread current directions nearly as many (arXiv 2512.10159)<sup>1</sup>. In op-amp circuits these slips show up as a feedback current sent the wrong way, which flips the sign of the gain, or as  $v_n$  set to 0 when the noninverting input is not grounded. S11 shows a direction slip, and S12 a formula used where it does not apply: find them before you trust any answer.
- Check every answer yourself: KCL at the inverting input must hold with your  $v_o$ , and  $v_o$  must lie between the supplies.
- The quiz is device-free, and every question is a twin of a practice-sheet question. If you can do the sheet on your own, you will do well on the quiz.

## Checklist: you can do these without help

- ☐ State the two ideal op-amp rules and when they apply
- ☐ Name an amplifier's configuration from its circuit and write  $v_o$  in terms of its inputs
- ☐ Analyse the inverting, summing, noninverting and difference amplifiers with KCL at the inverting input
- ☐ Find the range of an input that keeps the op amp linear, and the output when it saturates
- ☐ Find the current in a load and at the op amp's output
- ☐ Analyse two op amps in cascade and say which stage saturates first

---

<sup>1</sup><https://arxiv.org/html/2512.10159>